

SQL Practice Questions - Answers and Logic

Retrieve all columns from Employees

SQL Query:

```
SELECT * FROM Employees;
```

Logic:

The asterisk (*) returns every column in the table.

Retrieve EmployeeName and Salary

SQL Query:

```
SELECT EmployeeName, Salary FROM Employees;
```

Logic:

Specify only the columns needed instead of using *.

Employees earning above 50,000

SQL Query:

```
SELECT * FROM Employees WHERE Salary > 50000;
```

Logic:

WHERE filters rows before any grouping or aggregation occurs.

Customers from Lagos

SQL Query:

```
SELECT * FROM Customers WHERE City = 'Lagos';
```

Logic:

The = operator returns rows matching the exact value.

Order employees by salary descending

SQL Query:

```
SELECT * FROM Employees ORDER BY Salary DESC;
```

Logic:

DESC sorts from highest to lowest.

Products alphabetically

SQL Query:

```
SELECT * FROM Products ORDER BY ProductName ASC;
```

Logic:

ASC sorts from A to Z.

Count employees per department

SQL Query:

```
SELECT Department, COUNT(*) AS TotalEmployees FROM Employees GROUP BY Department;
```

Logic:

GROUP BY creates groups and COUNT counts rows within each group.

Total sales per sales agent

SQL Query:

```
SELECT SalesAgent, SUM(SalesAmount) AS TotalSales FROM Sales GROUP BY SalesAgent;
```

Logic:

SUM adds all sales amounts within each agent group.

Total employees

SQL Query:

```
SELECT COUNT(*) AS TotalEmployees FROM Employees;
```

Logic:

COUNT(*) counts every row in the table.

Count customers per city

SQL Query:

```
SELECT City, COUNT(*) AS TotalCustomers FROM Customers GROUP BY City;
```

Logic:

Groups customers by city and counts each group.

SQL Thinking Framework

Use WHERE to filter rows, GROUP BY to aggregate, HAVING to filter aggregated results, ORDER BY to sort, and JOINS to combine tables.